

Generative AI and the Rise of New Drug Modalities

Hojung Nam^{1,2}

¹Department of Pharmacy and Yonsei Institute of Pharmaceutical Sciences, College of Pharmacy, Yonsei University, 85 Songdogwahak-Ro, Yeonsu-Gu, Incheon, 21983, Republic of Korea

²Department of Integrative Biotechnology and Translational Medicine, Yonsei University, 85 Songdogwahak-Ro, Yeonsu-Gu, Incheon, 21983, Republic of Korea

Artificial intelligence is reshaping drug discovery, compressing candidate selection from years to months. This talk first reviews current achievements in AI-based drug discovery and introduces our generative models for de novo molecular design. The second part turns to modalities beyond conventional small molecules: models for RNA-binding molecule prediction and generation, which expand the druggable genome, and peptide drug discovery, where reinforcement-learning approaches enable species-aware antimicrobial peptide prediction and generation. Together, these efforts illustrate how generative AI is broadening both the chemical space and the modern drug development.